

OOPS ASSIGNMENT-6

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Q1. Implement a class Library that contains an array of Book objects. The Book class should have attributes like title, author, and ISBN.

CODE:

```
#include <iostream> using
```

```
namespace std;
```

```
class Book {
```

```
public:
```

```
    string title;
```

```
    string author;
```

```
    string ISBN;
```

```
};
```

```
class Library { Book
```

```
    books[100]; int
```

```
    count;
```

```
public:
```

```
    Library() { count
```

```
        = 0;
```

```
    }
```

```
    bool addNewBook(string &title, string &author, string &ISBN) {
```

```
        if (count >= 100) return false; books[count].title = title;
```

```
        books[count].author = author; books[count].ISBN = ISBN;
```

```
        count++;
```

```
        return true;
```

```
    }
```

```
    bool removeBooks(string &ISBN);
```

```

void displayDetails() {
    if (count == 0) {
        cout << "No books in
library\n"; return; }
    for (int i = 0; i < count; i++) {
        cout << "\nBook " << i + 1 << ":\n"; cout <<
        "Title: " << books[i].title << endl; cout <<
        "Author: " << books[i].author << endl; cout
        << "ISBN: " << books[i].ISBN << endl;
    }
}
};

```

```

bool Library::removeBooks(string &ISBN) {
    for (int i = 0; i < count; i++) {
        if (books[i].ISBN == ISBN) { for
            (int j = i; j < count - 1; j++) {
                books[j] = books[j + 1];
            } count--;
            return
            true;
        } }
    return
    false;
}

```

```

int main() { Library
    lib;

    string t1 = "Maths", a1 = "Rachit", i1 = "101";
    string t2 = "Data Structures", a2 = "Siya", i2 = "102";
    string t3 = "Analog", a3 = "Arun", i3 = "103"; string
    t4 = "DBMS", a4 = "Neha", i4 = "104"; string t5 =
    "OS", a5 = "Kirti", i5 = "105";

```

```
lib.addNewBook(t1, a1, i1);
lib.addNewBook(t2, a2, i2);
lib.addNewBook(t3, a3, i3);
lib.addNewBook(t4, a4, i4);
lib.addNewBook(t5, a5, i5);

cout << "\n Books after adding \n";
lib.displayDetails();          string
removeISBN    =    "103";    if
(lib.removeBooks(removeISBN))
    cout << "\nBook removed successfully\n";
else
    cout << "\nBook not found\n";

cout << "\n Books after removal \n";
lib.displayDetails(); return 0;
}
```

OUTPUT:

```

PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\" ; if (
    Books after adding

Book 1:
Title: Maths
Author: Rachit
ISBN: 101

Book 2:
Title: Data Structures
Author: Siya
ISBN: 102

Book 3:
Title: Analog
Author: Arun
ISBN: 103

Book 4:
Title: DBMS
Author: Neha
ISBN: 104

Book 5:
Title: OS
Author: Kirti
ISBN: 105

Book removed successfully

    Books after removal

Book 1:
Title: Maths
Author: Rachit
ISBN: 101

Book 2:
Title: Data Structures
Author: Siya
ISBN: 102

Book 3:
Title: DBMS
Author: Neha
ISBN: 104

Book 4:
Title: OS
Author: Kirti
ISBN: 105
PS D:\download\OOPS\Assignment6>

```

2Q. Implement the question 1 using constructor. Do the following modification:

- Add default constructor, parameterized constructor, and copy constructor to add the details of the book using the same parameters, i.e., title, author, and ISBN.
- Use the “this” pointer instead of scope resolution operator to access the data members of the class inside the constructor and function definition.
- Create function bool removeBooks(ISBN) same as the question 1.
- displayDetails() same as question 1.
- Create array of objects using 1._INITIALIZER list, 2. Dynamic initialization

CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
class Book {
```

```
public:
```

```
    string title;
```

```
    string author;
```

```
    string ISBN;
```

```
    Book() {
```

```
        title = ""; author
```

```
        = "";
```

```
        ISBN = "";
```

```
    }
```

```
    Book(string title, string author, string ISBN) {
```

```
        this->title = title; this->author = author;
```

```
        this->ISBN = ISBN;
```

```
    }
```

```
    Book(const Book &b) {
```

```
        this->title = b.title; this-
```

```
        >author = b.author;
```

```
        this->ISBN = b.ISBN;
```

```
    }
```

```
};
```

```
class Library { Book
```

```
    books[100];    int
```

```
    count;
```

```
public:
```

```
    Library() { count
```

```
        = 0;
```

```
    }
```

```
    bool addNewBook(string &title, string &author, string &ISBN) {
```

```

    if (count >= 100) return false; books[count]
    = Book(title, author, ISBN); count++;
    return true;
}

```

```

bool removeBooks(string &ISBN) { for
    (int i = 0; i < count; i++) {
        if (books[i].ISBN == ISBN) { for
            (int j = i; j < count - 1; j++) {
                books[j] = books[j + 1];
            } count--;
        } return
        true;
    }
} return
false;
}

```

```

void displayDetails() {
    if (count == 0) {
        cout << "No
        books\n"; return; }
    for (int i = 0; i < count; i++) {
        cout << "\nBook " << i + 1 << endl; cout <<
        "Title: " << books[i].title << endl; cout <<
        "Author: " << books[i].author << endl; cout <<
        "ISBN: " << books[i].ISBN << endl;
    }
}
};

```

```

int main() {
    Book arr1[3] = {
        Book("C++", "Ram", "101"),
        Book("DSA", "Shyam", "102"),
    }
}

```

```

    Book("OS", "Amit", "103")
};
cout << "Initializer list array:\n"; for
(int i = 0; i < 3; i++) {
    cout << arr1[i].title << " " << arr1[i].author << " " << arr1[i].ISBN << endl;
}

```

```

int n = 2;
Book *arr2 = new Book[n]; arr2[0] =
Book("DBMS", "Neha", "104");
arr2[1] = Book("Physics", "Kiran", "105");

```

```

cout << "\nDynamic array:\n"; for
(int i = 0; i < n; i++) {
    cout << arr2[i].title << " " << arr2[i].author << " " << arr2[i].ISBN << endl;
}

```

```

delete[] arr2; Library
lib;
string t1="Math", a1="A", i1="201";
string t2="Physics", a2="B", i2="202";
string t3="Chemistry", a3="C", i3="203";
string t4="Biology", a4="D", i4="204";
string t5="English", a5="E", i5="205";

```

```

lib.addNewBook(t1,a1,i1);
lib.addNewBook(t2,a2,i2);
lib.addNewBook(t3,a3,i3);
lib.addNewBook(t4,a4,i4);
lib.addNewBook(t5,a5,i5);

```

```

cout << "\nLibrary books:\n";
lib.displayDetails(); string
rem = "203";
lib.removeBooks(rem);

```

```

        cout << "\n    After removing one book: \n";

    lib.displayDetails();
    return 0;
}

```

OUTPUT:

```

PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\"
Initializer list array:
C++ Ram 101
DSA Shyam 102
OS Amit 103

Dynamic array:
DBMS Neha 104
Physics Kiran 105

Library books:

Book 1
Title: Math
Author: A
ISBN: 201

Book 2
Title: Physics
Author: B
ISBN: 202

Book 3
Title: Chemistry
Author: C
ISBN: 203

Book 4
Title: Biology
Author: D
ISBN: 204

Book 5
Title: English
Author: E
ISBN: 205

```

```

        After removing one book:

Book 1
Title: Math
Author: A
ISBN: 201

Book 2
Title: Physics
Author: B
ISBN: 202

Book 3
Title: Biology
Author: D
ISBN: 204

Book 4
Title: English
Author: E
ISBN: 205
PS D:\download\OOPS\Assignment6>

```


3Q. Create a class Account with attributes account number (of const long type), transaction ID (of long type), transaction type (credit/debit) (of string type), and balance (of double type).

CODE:

```
#include <iostream> using
namespace std;

class Account {
private:
    const long accNo;
    long      transID;
    string  transType;
    double balance;

public:
    Account(long accNo, long transID, string transType, double balance)
        : accNo(accNo)
    {
        this->transID = transID; this-
        >transType  =  transType;
        this->balance = balance;
    }
    void display() {
        cout << "Account No: " << accNo << endl; cout <<
        "Transaction ID: " << transID << endl; cout <<
        "Transaction Type: " << transType << endl; cout <<
        "Balance: " << balance << endl;
    }
};

int main() {
    Account a1(1001, 5001, "credit", 15000.5);
    a1.display();
    return 0;
}
```

OUTPUT:

```
Balance: 15000.5
PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\" ;
Account No: 1001
Transaction ID: 5001
Transaction Type: credit
Balance: 15000.5
PS D:\download\OOPS\Assignment6>
```

4Q. Write a program to add data objects of two different classes using friend functions.

CODE:

```
#include <iostream> using
namespace std;
```

```
class B;
```

```
class A { int
    x;
public:
    A(int x) { this->x
        = x;
    }
    friend int add(A, B);
};
```

```
class B { int
    y;
public:
    B(int y) { this->y
        = y;
    }
    friend int add(A, B);
};
```

```
int add(A obj1, B obj2) {
    return obj1.x + obj2.y;
}
int main() {
```

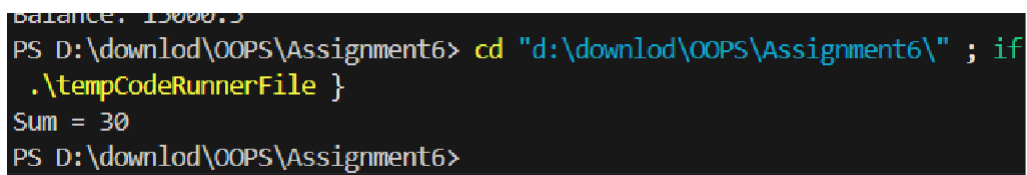
```

A obj1(10); B
obj2(20);
int result = add(obj1, obj2);
cout << "Sum = " << result << endl;

return 0;
}

```

OUTPUT:



```

Balance: 1000.5
PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\" ; if
.\tempCodeRunnerFile }
Sum = 30
PS D:\download\OOPS\Assignment6>

```

5Q. Define a class named Complex with properties (real and imaginary) and methods as per following details.

- Parameterized constructor and copy constructor to initialize object values.
- void display () to display complex number.
- void sum (Complex, Complex) to add two complex numbers (objects of Complex class). Implement this function as the friend function.

CODE:

```

#include <iostream> using
namespace std;

class Complex { int
    real, imag;
public:
    Complex(int r, int i) {
        real = r; imag = i;
    }
    Complex(const Complex &c) {
        real = c.real; imag = c.imag;
    }
    void display() {
        cout << real << " + " << imag << "i" << endl;
    }
}

```

```

    friend void sum(Complex, Complex);
};

void sum(Complex c1, Complex c2) { int r = c1.real +
    c2.real; int i = c1.imag + c2.imag; cout << "Sum =
    " << r << " + " << i << "i" << endl;
}

int main() {
    Complex c1(2, 3);
    Complex c2(4, 5); Complex
    c3(c1); cout << "First
    complex: "; c1.display();
    cout << "Second complex: ";
    c2.display(); cout <<
    "Copied complex: ";
    c3.display();
    sum(c1, c2);

    return 0;
}

```

OUTPUT:

```

PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\" ;
.\tempCodeRunnerFile }
First complex: 2 + 3i
Second complex: 4 + 5i
Copied complex: 2 + 3i
Sum = 6 + 8i
PS D:\download\OOPS\Assignment6>

```